

General and Structural Chemistry

Quiz-1

Spring Semester 2026

IIIT-Hyderabad

Time: 45 min

Max Marks: 45

Q1. State the following statements as True or False:

[5×2]

- (i) The function $\sin kx$ is an eigenfunction of the one-dimensional kinetic energy operator.
- (ii) The radial functions $R_{n,l}$ vanishes at the nucleus for all allowed values of n and l .
- (iii) The value of the integral $\int (-\psi_{2,1,1})^* (\psi_{2,1,-1}) dV$ (integrated over the whole space) is 1.
- (iv) An electron in an orbital with $n = 1$ in a hydrogen atom has lower energy than a free electron ($n = \infty$).
- (v) Particles with half-integer spin are called bosons.

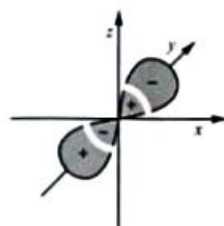
Q2. What is the magnitude of the total angular momentum for the wavefunction $\psi_{2,1,0}$? What is the magnitude of its z-component? [2,2]

Q3. How do we obtain the relation: $0 \leq l \leq (n - 1)$, where $n = 1, 2, 3, \dots, \infty$? [4]

Q4. When l gets very large, there is no quantization which implies angular momentum values become continuous. Explain the above observation. [2]

Q5. Show that the $2p_x$ orbital has a nodal plane in the y-z plane. [3]

Q6. (i) Identify the following orbital. Justify your answer. [1,2]



(ii) Comment whether the above orbital has gerade or ungerade symmetry with respect to inversion. [1]

(iii) How many orbitals of this type are found in a shell with $n = 9$? [1]

(iv) What is the smallest possible n value for an orbital of this type? [1]

Q7. Give schematic diagrams of the radial function $R(r)$ and the radial distribution function $|R_{n,l}|^2 r^2$ for orbitals in the hydrogen atom for $n = 3$. Draw all the radial functions $R(r)$ under the same x-y axis. Similarly, under a separate x-y axis, show all the plots of $|R_{n,l}|^2 r^2$. [3,3]

Q8. Define an orbital. What are the four major information one can obtain from an orbital? [1,2]

Q9. What is the most probable radius for an electron in the 1s orbital of hydrogen? Given: 1s orbital: $\psi_{1s}(n, l, \phi) = \frac{1}{\sqrt{2}} \left(\frac{1}{a_0}\right)^{3/2} e^{-\frac{r}{a_0}}$ [3]

Q10. How many components did the beam of silver atoms passing through an inhomogeneous magnetic field split into in the Stern-Gerlach experiment? What can one conclude about the magnitude of the spin quantum number from this observation? Justify your answer using the results of angular orbital quantum number. What is the magnitude of the total spin angular momentum? [1,2,1]

.....

Info:

$$p_z = (3/4\pi)^{1/2} \cos\theta \quad f(r) \cdot r \sin\theta \cos\phi = 2p_x$$

$$\tilde{L}^2 = \tilde{L}_x^2 + \tilde{L}_y^2 + \tilde{L}_z^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right]$$

$$R_{n,l}(r) = -\sqrt{\left(\frac{2Z}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}} \cdot \left(\frac{2Zr}{na_0}\right)^l \cdot e^{-Zr/na_0} \cdot L_{n+l}^{2l+1}\left(\frac{2Zr}{na_0}\right)$$

.....